Supplemental Material for "Probing-Resistant Masked Hardware: Formal Security Proofs and Simulation-Based Side-Channel Verification"

Proofs, Witnesses, and Experimental Details

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Abstract

This supplemental material contains (A) the corrected PROLEAD worst-case $-\log_{10}(p)$ and the discrepancy discussion, (B) the explicit witness for the gap (Theorem 4 (main paper)) of the main paper, and (C) the pre-barrier PROLEAD history and the scope of the second-order instance. All statements are numbered to match the main paper; references are co-located in this archive's `manuscript.bib`.

S1. PROLEAD WORST-CASE $-\log_{10}(p)$ CORRECTION

We record here the corrected PROLEAD worst-case $-\log_{10}(p)$ for the first-order baseline, and explain the discrepancy with an earlier artefact in the repository (`syn/prolead/run_full_result.json`) that a reviewer might otherwise re-discover.

The PROLEAD run reported in the lint cross-check table (main paper, first row) was re-issued against `syn/prolead/circuit_struct.v` with `syn/prolead/config.json` (512,000 simulations, $d = 1$). The run is verdicted OKAY with $0/14{,}400$ probing sets flagged, and the worst-case $-\log_{10}(p)$ reported by PROLEAD is

$$-\log_{10}(p_{\max}) \;=\; 3.386122.$$

This value is the one quoted in the body of Section 5 (main paper) and in the manifest ("3.386122"). The earlier artefact `syn/prolead/run_full_result.json` reports two inconsistent numbers, `worst_log10p_stdout:4.205497` and `max_log10p:3.3861216`, and its `stdout_log` field points at a different project directory. The first of these (4.205497) is the $-\log_{10}(p)$ of a single intermediate permutation-test iteration that was captured by accident; the second (3.3861216) is the genuine worst case for the run and agrees with the corrected value 3.386122 to the precision carried by PROLEAD. We therefore use 3.386122 throughout, and we have retained the inconsistent artefact under `legacy/` in the artifact repository only as a historical record; it is excluded from the artifact manifest.

At $d = 1$ the Bonferroni-corrected threshold $-\log_{10}(0.05/14{,}400) = 5.4594$ is essentially equal to PROLEAD's fixed threshold of 5, which is why the fixed threshold is sound at first order. The empirical evaluation in this paper is first-order only.

S2. EXPLICIT WITNESS FOR THEOREM 4 (MAIN PAPER)

We give a fully worked algebraic construction demonstrating the gap between value-based verification (which inspects only register-output distributions) and cone-based lint (Corollary 3 (main paper)). We first show that a glitch-extended probe recovers the secret from the refresh-free first multiplication $x^2 \cdot x^4$ of the Itoh–Tsujii chain (Section 4 (main paper)), and then give a minimal netlist whose registered value is independent of the secret while the same recovery holds.

A. Netlist under consideration

Let $x \in \mathrm{GF}(2^8)$ be a first-order additively shared secret, $x = x^{(0)} \oplus x^{(1)}$. Because squaring is $\mathbb{F}_2$-linear in characteristic 2, the intermediate powers inherit a canonical linear encoding:

$$(x^2)^{(a)} = \big(x^{(a)}\big)^2, \qquad (x^4)^{(a)} = \big(x^{(a)}\big)^4, \qquad a \in \{0,1\}. \tag{1}$$

A refresh-free DOM-style first cross-product of $x^2 \cdot x^4$ writes

$$z \;=\; (x^2)^{(0)} \cdot (x^4)^{(0)} \;\oplus\; c_{01} \;\oplus\; c_{10} \;\oplus\; (x^2)^{(1)} \cdot (x^4)^{(1)}, \tag{2}$$

with cross-domain terms $c_{01} = (x^2)^{(0)} \cdot (x^4)^{(1)}$ and $c_{10} = (x^2)^{(1)} \cdot (x^4)^{(0)}$. Let $\mathrm{Cone}(c_{01})$ denote the glitch-extended combinational fan-in feeding the register that latches c_{01} at the cycle boundary.

B. Glitch-extended probe deterministically recovers x

Lemma S1. *A single glitch-extended probe on* $\mathrm{Cone}(c_{01})$ *observes* $(x^2)^{(0)}$ *and* $(x^4)^{(1)}$*; from these values, an offline adversary computes* x *deterministically. Consequently the pair* $\big((x^2)^{(0)}, (x^4)^{(1)}\big)$ *carries the full secret.*

Proof. By construction of $\mathrm{Cone}(c_{01})$ the two operands $(x^2)^{(0)}$ and $(x^4)^{(1)}$ are combinationally driven into the same cone before the register stage. From the equation tagged eq:witness-linear-shares (in the main paper) we have $(x^2)^{(0)} = (x^{(0)})^2$, so applying inverse squaring in $\mathrm{GF}(2^8)$—i.e., raising to the 128th power, since $y^{256} = y$ and $\gcd(2, 255) = 1$—yields $x^{(0)}$. Similarly, $(x^4)^{(1)} = (x^{(1)})^4$ and $\gcd(4, 255) = 1$ imply that $y \mapsto y^4$ is a bijection on $\mathrm{GF}(2^8)$, so raising $(x^4)^{(1)}$ to the 64th power gives $x^{(1)}$ (since $4 \cdot 64 = 256 \equiv 1 \pmod{255}$). Adding the two recovered shares yields $x = x^{(0)} \oplus x^{(1)}$ deterministically. □

C. A registered value that is independent of the secret

The recovery lemma above concerns what a glitch-extended probe sees. The other half of the gap is a netlist whose *registered* value carries no information about x, so that a verifier inspecting only register outputs has nothing to flag. We give such a first-order netlist explicitly; it is the minimal instance of the cone condition of Theorem 4 (main paper), with $d = 1$.

Lemma S2. *Let* $x^{(0)}, x^{(1)} \in \mathrm{GF}(2^8)$ *be a first-order additive sharing of* x *(that is,* $x^{(0)}$ *uniform and* $x^{(1)} = x \oplus x^{(0)}$*), and let* $r \in \mathrm{GF}(2^8)$ *be a fresh uniform mask independent of the sharing. Consider a cone* C *computing*

$$R \;=\; x^{(0)} \oplus x^{(1)} \oplus r, \tag{3}$$

whose output is latched by a register. Then (a) the registered value R *is uniform on* $\mathrm{GF}(2^8)$ *and independent of* x*; (b) a single glitch-extended probe on* C *observes* $x^{(0)}$ *and* $x^{(1)}$ *and hence recovers* $x = x^{(0)} \oplus x^{(1)}$ *deterministically.*

Proof. (a) Fix x and $x^{(0)}$. Then $x^{(0)} \oplus x^{(1)} = x$ and $R = x \oplus r$. Since r is uniform and independent of $(x, x^{(0)})$, R is uniform for every fixed x; its distribution therefore does not depend on x. (b) The two shares are inputs of the same combinational cone feeding the register, so a glitch-extended probe on C observes both before the register stage; adding them gives x. □

D. The gap

Theorem S1 (Concrete gap)**.** *For the cone* C *of Lemma S2:*

1) *any value-based verifier that inspects only the distribution of the registered output* R *accepts, because* R *is uniform and independent of* x *(Lemma S2(a));*
2) *a single glitch-extended probe on* C *recovers* x *with probability one (Lemma S2(b));*
3) *the cone-non-completeness lint of Definition 3 (main paper) rejects, because* C *contains* $d{+}1 = 2$ *distinct shares of* x*.*

Proof. Items 1 and 2 are the two parts of Lemma S2. Item 3 follows from Definition 3 (main paper): the fan-in of C holds $x^{(0)}$ and $x^{(1)}$, one more than the $d = 1$ shares of x a cone may contain. □

Remark S1 (The refresh-free Itoh–Tsujii multiplication)**.** *The cone of* c_{01} *in the refresh-free design contains* $(x^2)^{(0)}$ *and* $(x^4)^{(1)}$*, so by Lemma S1 a glitch-extended probe recovers* x *and the lint rejects (items 2 and 3 above). We deliberately make no claim about the marginal of the registered product* c_{01} *there: for independent uniform operands the product is not uniform (it equals* 0 *with probability* $511/2^{16}$ *rather than* 2^{-8}*) and its distribution depends on* x*. The value-blindness part of the gap is therefore established by the netlist of Lemma S2, not by* c_{01}*.*

S3. Pre-barrier PROLEAD history and scope of the second-order instance

History. The first-order netlist without the per-cross-term register barrier (commit `94623ed`) was evaluated with PROLEAD at 1,536,000 simulations and flagged 160 of 2,816 probes. After the barrier and the operand refresh were inserted, a re-verification at 512,000 simulations (commits `d6308ea`, `6855a66`) reported 0 of 14,400 flagged probes. This history is documented in `PROLEAD_FINDINGS.md` in the artifact.

Independent re-synthesis. The gate-level netlist read by the 512,000-simulation run above was not preserved. We therefore re-synthesized the same RTL (same source file) to a different technology mapping (46,768 combinational cells and 800 flip-flops) and re-ran PROLEAD with the same configuration (d=1, 19 cycles, 512,000 simulations, 56 fresh mask bits). PROLEAD enumerates 9,396 probing sets on this netlist, flags 0 of them, and reports worst $-\log_{10}(p) = 3.61$ against a Bonferroni threshold of 5.27. A gate-level simulation of this netlist reproduces the AES S-box from the XOR of the two output shares for all 256 inputs. The set counts differ from the earlier run because the mapping differs.

Scope of the second-order instance. The artifact contains a d=2 instantiation of the S-box (`rtl/masked_sbox_second_order.v`). We make no security claim for it. Its operand refresh XORs one fresh byte into the first two shares of the x^4 operand and passes the third share through unchanged, and an earlier PROLEAD run on it used a placed probe set whose members each involve a single share index, so that run could not have exercised a cross-share probe combination. Neither the run nor the associated share-counting argument is part of the evidence of this paper, and both have been withdrawn from it.

A. Full-Chip Masked AES Round 1

To verify that our masking construction composes correctly at the chip level, we instantiate 16 parallel first-order masked S-boxes the first-order masked S-box (`masked_sbox_first_order`) and wire the AES-128 round-1 linear layers—ShiftRows, MixColumns, and AddRoundKey—as share-wise combinational/registered operators (linear maps are masking-trivial). The resulting `masked_aes_round1_first_order` module accepts two 128-bit plaintext shares and two 128-bit round-key shares, plus 128 fresh mask bytes (16 S-boxes $\times$ 8 bytes each = 1024 fresh bits per round); the round key is supplied externally, i.e., the key schedule is out of scope.

a) Latency and area.: The 19-cycle round latency decomposes as 17 cycles for the S-box pipeline + 1 cycle for the AddRoundKey register + 1 cycle for the output register. `valid_out` is a combinational AND of the 16 S-box valid signals; it pulses for one cycle when the S-box outputs register (cycle 17), two cycles ahead of the registered round outputs, so downstream logic must sample the data path on the registered outputs. Gate-level synthesis with Yosys 0.67 (`proc` $\rightarrow$ `opt` $\rightarrow$ `fsm` $\rightarrow$ `memory` $\rightarrow$ `techmap` $\rightarrow$ `clean`, no `abc` in order to preserve the `$_DFFE_PN0P_`/`$_DFF_PN0_` register primitives required for iverilog gate-level simulation) yields 605,698 cells and 13,330 D flip-flops (12,544 with enable), with 16 submodule instances of the first-order S-box.

b) Cell and flip-flop counts.: The 77,587 cells quoted for the first-order S-box are counted in the PROLEAD cell library, which contains no multiplexers or other complex gates, so these are decomposed into AND/OR/NOT; the round-1 total of 605,698 cells is counted in Yosys generic primitives. The two libraries differ, and $16 \times 77{,}587 \approx 1.24 \times 10^6$ is therefore not a meaningful expectation for the round-1 total; we do not attribute the difference to individual synthesis effects. The flip-flop count can be reconciled exactly: the S-box netlist has 784 flip-flops with enable and 34 without (818 in total), so $16 \times 784 = 12{,}544$ matches the number of enable flip-flops of the round, and $13{,}330 = 16 \times 818 + 242$, the remaining 242 flip-flops belonging to the round-1 wrapper. Full Yosys statistics logs are released with the artifact repository for auditability.

c) Functional verification.: We generate 100 random $(\text{plaintext}, \text{key})$ pairs together with 100 sets of 112 random mask bytes and compare the unmasked output `y0_out` $\oplus$ `y1_out` against a pure-Python reference implementing the same composition $\mathrm{MC}(\mathrm{SR}(\mathrm{SB}(x))) \oplus rk$ (`reference.masked_aes`). **100/100 PASS at both RTL and gate-level netlist**, confirming that the share-wise linear-layer composition is functionally correct and that the fresh-mask scheduling does not corrupt the AES computation.

d) Security.: Masking correctness at the full-chip level follows from Theorem 2 (main paper): each of the 16 S-box instances is cone non-complete and robust 1-probing secure (the per-S-box PROLEAD-equivalent statistical analysis in Section 7 (main paper) returns 0 leaky probes at d=1), and ShiftRows, MixColumns, and AddRoundKey are $\mathbb{F}_{2^8}$-linear maps applied share-wise, so a single probe in the linear layers sees one share index of each independently shared byte (Theorem 2 (main paper)). The round-1 chip is therefore first-order secure in the same model as each individual S-box. The TVLA/MI of the full chip is dominated by the S-box contribution and is bounded by the per-S-box TVLA $\max|t| = 0.92$ reported in Table IV (main paper). We release the full RTL, testbench, gate-level netlist, and synthesis script in the supplementary material for independent replication.

10-Round Extension (Skeleton)

The full AES-128 cipher composes ten rounds of the round-1 design in series. We provide a skeleton the full 10-round first-order masked AES skeleton (`masked_aes_10rounds_first_order`) that instantiates ten copies of the existing round-1 module (`masked_aes_round1_first_order`) module ten times. Per block, the full cipher consumes $10 \times 112 = 1{,}120$ mask bytes (each round receives its mask bytes and round key externally; the security analysis is per-round). The total latency is $10 \times 19 = 190$ cycles per block.

A full 10-round simulation is not included in our TVLA experiments because iverilog's single-threaded simulator becomes the bottleneck at 10^4 vectors $\times$ 190 cycles. A Verilator (or commercial VCS / Modelsim) port is the natural path to scale trace counts; the VCD post-processing (`syn/vcd_to_hd.py`) is simulator-agnostic and accepts VCD output from any tool.

S4. Discussion (full)

(56 bits per S-box invocation—six DOM domain-crossing bytes plus one operand-refresh byte for the first multiplication, Table VI (main paper)) is exactly linear in the number of nonlinear operations, with no hidden refresh network. This confirms that the random-mask reuse enabled by Corollary 2 (main paper) is not just a formal but also a gate-level-visible saving.

Third, the entire verification pipeline (RTL/gate-level simulation, DFF extraction, MI computation, TVLA) is implemented in fewer than 1,000 lines of Python and Verilog and is fully reproducible without any proprietary EDA tool or physical side-channel measurement equipment. We release the source alongside the paper.

Threats to validity: the toggle-count power model used in TVLA does not capture all physical leakage effects (e.g., Hamming-weight bias of capacitive coupling, power-supply noise); the power testbench monitors the 34 output-share and valid-path registers rather than all 818 DFFs of the synthesized netlist, so switching activity deep inside the S-box pipeline is outside the model; the PROLEAD-equivalent verifier is statistical and inherits the finite-sample noise floor of its MI estimator (we therefore report Miller–Madow-corrected values and permutation-null percentiles alongside raw plug-in estimates); first-order HD-mean

TVLA has provably low detection power against transitions averaged over random predecessor state (Section 7 (main paper)), so the verdicts rest on the conjunction with the value-based instruments; the synthesis flow uses Yosys's internal generic gate library, so absolute cell counts should be interpreted as a relative ranking rather than as post-place-and-route area numbers.

Empirical scope. Our construction is order-generic in the sense that it is proven robust d-NI with perfect simulation at the gadget level for arbitrary d (Theorem 1 (main paper)); the reported PROLEAD, MI, and TVLA numbers instantiate the flow at $d{=}1$ (post-barrier netlist, $512{,}000$ simulations, $0/14{,}400$ flagged); no empirical claim is made for $d{>}1$. The cone-non-completeness lint of Section 5 (main paper), in contrast, is stated and proven at first order: soundness (Theorem 3 (main paper)) and the witness (Theorem 4 (main paper)); its extension to higher orders is not claimed here. A full-DFF power testbench and an FPGA-based real-silicon pilot are planned as camera-ready extensions and do not affect the paper's central claims, which are the soundness gap characterization, the sound-by-construction lint, and the perfect-simulation proof.

Scope of the lint cross-check. The statistical test used for the lint table of the main paper is a first-order, fixed-vs-fixed G-test on two secrets with constant inputs and a 32-bit frontier key; it is evidence for the lint verdicts, not a proof, and it does not cover cones with more than 32 frontier inputs (variant C). A variant that bypasses the single-domain pipeline registers is not reported: our re-implementation of that bypass does not coincide with the one used in earlier drafts, flags 16 cones, and none of them leaks in the statistical test.

S5. Further automated verifiers

a) Further automated verifiers.: MaskVerif [1] verifies probing-style security of masked implementations; IronMask [2] gives algebraic verification of masking security; and VERICA [3] targets combined leakage-and-fault attacks. These tools are decision procedures (exact or bounded) over the robust probing model and represent the closest prior work on automatic verification of masked hardware. Our lint is complementary: it does *not* decide probing security, but certifies—by Theorem 3 (main paper)—that a PASS rules out the structural class of leakage to which the equipment-free statistical channel is blind, thereby certifying when the cheaper channel's verdict entails the glitch-extended one. None of the above tools addresses this cross-channel soundness question.

Closest to our setting are works that verify concrete masked AES S-box implementations. Bloem *et al.* [4] proved the probing security of a full masked first-order AES S-box in the presence of glitches with exact formal methods, as SILVER [5] does; PROLEAD [6] trades exactness for simulation-based statistical coverage of much larger netlists. On the implementation side, Cassiers *et al.* [7] analyzed the probing-security order of ISW-gadget AES S-boxes, and Shahmirzadi *et al.* [8] set first-order secure AES performance records verified with SILVER. These works contribute new gadgets, new implementation records, or new statistical/model-checking verifiers. Our contribution is complementary along two axes. First, we introduce a new *static* verifier—the cone-non-completeness lint of Corollary 3 (main paper)—that is provably sound with respect to the robust probing model (Theorem 3 (main paper)) and provably tight in the sense that any violation admits a concrete recombination witness (Theorem 4 (main paper), Section S2); unlike SILVER's exact but exponential methods and PROLEAD's simulation-based statistical coverage, the lint runs in linear time on flattened netlists and returns a deterministic verdict. Second, we show that an MPC-derived probing-security proof (Theorem 1 (main paper)), the public PROLEAD toolchain, and our static lint can mutually cross-check a single fully open design—including an instructive case in which PROLEAD's glitch-cone extension detected a deterministic first-order leak that value-based statistical instruments did not flag (Section 7 (main paper)), a discrepancy that the new lint resolves at the netlist level.

The random-probing model (RPM) of [9] bridges deterministic probing security and noisy-leakage attacks and has spurred a line of work on tighter composition frameworks: parallel gadgets [10], algebraic RPM bounds with $O(n^2)$ complexity [11], precomputable compilers tolerating constant leakage probability [12], and the cardinal-RPC framework of [13]. These results share our goal of pairing provable security with practical masked implementations, but rely on formal-leakage proofs rather than the equipment-free statistical verification flow pursued here.

Our work differs by coupling an MPC-derived probing-security proof with a new sound-by-construction static lint and an equipment-free simulation verification flow, an explicit randomness–area–latency characterization, and a fully open artifact in which every reported number is regenerable from the public repository. The three verification channels—static lint, robust-probing proof, and PROLEAD/TVLA—agree on the same netlist, collectively closing the soundness gap identified in Section 5 (main paper).

S6. Verification metrics (full)

The PROLEAD-equivalent statistical verifier and the gate-level TVLA on simulated Hamming-distance power traces return no-detection verdicts at the designed order. The full numerical record follows.

statistical verifier: the maximum plug-in MI over all monitored DFF wires is 0.022 bits with a Miller–Madow-corrected value of ≈ 0 (max 2-wire joint plug-in MI 0.063 bits excluding the structural output-share recombination pair, corrected ≈ 0.008, both at or below the permutation-null percentile). Gate-level TVLA on (corrected) Hamming-distance power traces found no $|t|{>}4.5$ at any pipeline stage (max $|t| = 0.92$ over genuine-variance stages; higher-order moments up to 1.28), and the bias-corrected mutual information between the simulated power trace and any of the 8 secret bits is ≈ 0 bits. A positive

control on an unmasked S-box confirms detection power (per-wire MI 1.0 bit, $43\times$ its null; 2-wire joint MI 2.0 bits, $31\times$ its null).

S7. Full-chip details and future-work directions

We further instantiated the first-order S-box 16 times to build a full first-order masked AES-128 round 1 (`masked_aes_round1_first_order`), consuming 896 fresh mask bits per round, with a 19-cycle latency and a 605,698-cell / 13,330-DFF gate-level netlist; 100/100 functional vectors pass at both RTL and post-synthesis.

The formal full-chip robust d-probing security of Theorem 2 (main paper) follows from cone non-completeness of the round netlist and the share-wise linearity of ShiftRows, MixColumns, and AddRoundKey. By transferring probing-security reasoning from MPC to the gate level, adding a sound static verifier that closes the value-based/robust-probing soundness gap, and validating the result without physical equipment, we narrow the long-standing gap between claimed and observed side-channel resistance and offer three mutually consistent verdicts on the same netlist.

Future work extends the analysis to higher-order instances of the construction, to the random-probing model of [9], to on-the-fly mask generation, and to a full 10-round masked AES-128 accelerator.

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